
Data Structures

BS (CS) _Fall_2024

Lab_12 Tasks



Learning Objectives:

1. AVL Tree

Lab Tasks

Submission Instructions

1. Create a separate **.h** and **.cpp** file for each class.
2. Now create a new folder with name *ROLLNO_SEC_LAB12* e.g. **i23XXXX_A_LAB07**(Also name your **cpp** file like this.)
3. Move all of your files to this newly created directory and compress it into a **.zip file**. Return the test file as well.
4. Now you have to submit this zipped file on Google Classroom.
5. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.
6. Plagiarism or AI generated code in the Lab Task will result in **zero** marks in the whole category.

Question 1

Write a function that performs rotations on a binary search tree depending upon the key value of the node

- a. If key is a prime number, make no rotation
- b. If key is even (and not a prime) making left rotation
- c. If key is odd (and not a prime) make right rotation

At the end display the resultant tree

Note: you must handle all cases

Question no. 2

Suppose you have an AVL tree containing distinct integer keys. Design an algorithm to split this AVL tree into two separate AVL trees, TreeA and TreeB, such that TreeA contains keys less than or equal to a given threshold value K, and TreeB contains keys greater than K. The algorithm should ensure that both resulting trees maintain the AVL property.